

Alexandre Sauquet

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EDUCATION

Purdue University West Lafayette, IN
Bachelor of Science: Computer Science Honors, Artificial Intelligence, Mathematics, Statistics August 2023 — May 2027

- Cumulative GPA: 3.88/4.00
- Data Mining & Machine Learning, Natural Language Processing (Graduate), Reinforcement Learning (Graduate), Analysis of Algorithms, Theory of Computation and Computational Complexity (Graduate), Statistical Theory, Linear Algebra II, Computational Methods in Optimization (Graduate), Numerical Methods, Elementary Stochastic Processes.

PROFESSIONAL EXPERIENCE

Google | A/B testing, Java May 2026 — Present
Software Engineering Intern Sunnyvale, CA

- Designed and configured controlled A/B experiments to evaluate promotional strategies for Gmail in Workspace, defining treatment and control arms within a production ML-based ranking system serving millions of business users.
- Extending the same experimental-design methodology to a second study on Google Voice promotions, measuring lift across the full business and enterprise user base.

CAST Software | FastMCP, Neo4j May 2025 — August 2025
Intern AI New York City, NY

- Bridged CAST's code-analysis platform to AI agents in GitHub Copilot and VS Code by developing 44 MCP tool APIs spanning 150+ languages/frameworks, collapsing typical analysis workflows from 20+ UI steps to a single natural-language prompt.
- Built the integration's first deployment path by packaging the MCP server runtime as a Docker container, enabling single-command rollout to enterprise customers.

Duarte, Inc. | HuggingFace, LlamaIndex May 2024 — August 2024
Jr. AI Engineer Intern Santa Clara, CA

- Implemented advanced AI models within consulting software solutions by creating custom pipelines, generating empathic human-like presentations that increased production speed by 29%.
- Architected software leveraging large multimodal models that analyzed diverse user-uploaded content with safe guardrails, expanding the company's service capabilities.

Purdue University Research | PyTorch, Optimization Theory November 2023 — May 2026
Undergraduate Researcher West Lafayette, IN

- Derived global convergence proofs for gradient descent in the Edge of Stability regime on 2D regularized logistic regression under Prof. Ruizhe Zhang, characterizing dynamics at step sizes beyond the classical stability threshold, with higher-dimensional extensions.
- Built the consultation agent for MAIA, a multi-agent voice assistant expanding rural healthcare access, orchestrating task-specific LLMs in a phone-based pipeline (Twilio speech-to-text, DeepGram voice synthesis) that runs without broadband so patients in low-connectivity areas can reach care by phone.

PROJECTS

TE Connectivity AI Cup: 1st place National, AI Assisted Manual Assembly Process Analysis Dec 2023 — May 2024

- Achieved 1st place Nationally and 2nd place Internationally in TE Connectivity's Collegiate Competition by outperforming 46 teams from 31 universities.
- Trained and optimized a diffusion transformer model for manufacturing action segmentation by fine-tuning hyperparameters, achieving 88.7% accuracy in production environment analysis.
- Constructed a patented hand detection system using YOLO architecture by implementing automatic task labeling, enabling pipeline generalization across multiple production lines.
- Generated \$170,000 in revenue by securing over 50 deployment opportunities, delivering a complete return on investment within 3 months.

SKILLS

- Programming Languages:** C++, Python [PyTorch, TensorFlow, Keras, OpenCV, Matplotlib, sklearn, Pandas, Metalog, NumPy, SciPy, Ultralytics, FastMCP], Java, C, JavaScript, x86_64, Julia, Bash.
- Technologies:** React, Git, GitHub, UNIX/Linux, CVAT, DagsHub, Typst, LaTeX, SLURM, Neo4j, QuantConnect, LlamaIndex.
- Languages:** French (native), English (bilingual).